Comprehensive Excel Solution Guide: ChurnQuest Telecommunications Churn Analytics

Target Document: Airtel Telecommunications Data Case Study — *ChurnQuest*

Format: Excel Formula Implementation & Analytical Guide

Data Setup & Structural Standards

Standardize your raw dataset sheet named **`Data`** with data starting at Row 2 (Headers in Row 1):

Column	Header Name	Data Type / Description
A	`state`	Text: 2-letter US state code
B	`account_length`	Integer: Number of months customer has been with telco
C	`area_code`	Text: Format `area_code_AAA` (e.g., `area_code_415`)
D	`international_plan`	Text: Binary (`yes` / `no`)
E	`voice_mail_plan`	Text: Binary (`yes` / `no`)
F	`number_vmail_messages`	Integer: Count of voicemail messages
G	`total_day_minutes`	Continuous: Day call duration in minutes
H	`total_day_calls`	Integer: Number of day calls
I	`total_day_charge`	Continuous: Day call charges ($)
J	`total_eve_minutes`	Continuous: Evening call duration in minutes
K	`total_eve_calls`	Integer: Number of evening calls

Column	Header Name	Data Type / Description
L	`total_eve_charge`	Continuous: Evening call charges ($)
M	`total_night_minutes`	Continuous: Night call duration in minutes
N	`total_night_calls`	Integer: Number of night calls
O	`total_night_charge`	Continuous: Night call charges ($)
P	`total_intl_minutes`	Continuous: International call duration in minutes
Q	`total_intl_calls`	Integer: Number of international calls
R	`total_intl_charge`	Continuous: International call charges ($)
S	`number_customer_service_calls`	Integer: Support calls logged
T	`churn`	Text: Target variable (`yes` / `no`)

Calculated Helper Columns (Add to `Data` Sheet)

Add these calculated columns to simplify downstream formula construction:

1. **`is_churn` (Column U):** Binary churn flag (`1` for `yes`, `0` for `no`).

   ```excel
   =IF(T2="yes", 1, 0)
   ```

2. **`is_intl_plan` (Column V):** Binary indicator for international plan (`1` for `yes`, `0` for `no`).

   ```excel
   =IF(D2="yes", 1, 0)
   ```

3. **`is_vmail_plan` (Column W):** Binary indicator for voicemail plan (`1` for `yes`, `0` for `no`).

   ```excel
   ```

```
=IF(E2="yes", 1, 0)
```

4. `total_monthly_charge` **(Column X):** Sum of daily, evening, night, and international charges.

```excel
=SUM(I2, L2, O2, R2)
```

Part 1: Basic-Level Questions (Q1 - Q10)

Question 1: Count and Proportion of Churned vs. Non-Churned Customers

- **Objective:** Calculate overall churn volume and churn rate percentage.
- **Excel Summary Formulas:**
  - **Total Customer Count (N):**

    ```excel
    =COUNTA(Data!T2:T4251)
    ```
  - **Churned Customers Count:**

    ```excel
    =COUNTIF(Data!T2:T4251, "yes")
    ```
  - **Retained Customers Count:**

    ```excel
    =COUNTIF(Data!T2:T4251, "no")
    ```
  - **Overall Churn Rate (%):**

    ```excel
    =COUNTIF(Data!T2:T4251, "yes") / COUNTA(Data!T2:T4251)
    ```
    (Format cell as Percentage `0.00%` → Baseline `$\approx 14.07\%$`)

Question 2: Average Account Length Comparison by Churn Status

- **Objective:** Compare average account length (`account_length`) between churned and retained customers.
- **Excel Summary Formulas:**
    - **Average Tenure (Retained):**

```excel
=AVERAGEIF(Data!T2:T4251, "no", Data!B2:B4251)
```

    - **Average Tenure (Churned):**

```excel
=AVERAGEIF(Data!T2:T4251, "yes", Data!B2:B4251)
```

    - **Tenure Difference (Delta):**

```excel
=AVERAGEIF(Data!T2:T4251, "yes", Data!B2:B4251) -
AVERAGEIF(Data!T2:T4251, "no", Data!B2:B4251)
```

- **Business Insight:** Mean tenure is nearly identical (~100.3 vs. ~102.7 months). Tenure alone does not buffer against churn; customer attrition occurs across all tenure brackets.

Question 3: Subscription to International Plans and Churn Correlation

- **Objective:** Evaluate how active international plans influence churn probability.
- **Excel Summary Formulas:**
    - **Churn Rate (No International Plan):**

```excel
=AVERAGEIFS(Data!U2:U4251, Data!D2:D4251, "no")
```

    - **Churn Rate (Has International Plan):**

```excel
=AVERAGEIFS(Data!U2:U4251, Data!D2:D4251, "yes")
```

- **Relative Churn Risk Factor:**

```excel
=AVERAGEIFS(Data!U2:U4251, Data!D2:D4251, "yes") /
AVERAGEIFS(Data!U2:U4251, Data!D2:D4251, "no")
```

- **Business Insight:** Customers with an international plan churn at more than **double** the rate (~29.25% vs ~12.21%). Mismatched pricing or unexpected fees drive cancellation.

Question 4: Customer Service Calls Comparison

- **Objective:** Determine if churned customers make significantly more support calls.
- **Excel Summary Formulas:**
  - **Average Support Calls (Retained):**

```excel
=AVERAGEIF(Data!T2:T4251, "no", Data!S2:S4251)
```

  - **Average Support Calls (Churned):**

```excel
=AVERAGEIF(Data!T2:T4251, "yes", Data!S2:S4251)
```

  - **Call Volume Increase:**

```excel
=AVERAGEIF(Data!T2:T4251, "yes", Data!S2:S4251) -
AVERAGEIF(Data!T2:T4251, "no", Data!S2:S4251)
```

- **Business Insight:** Churned customers make significantly more support calls (~2.23 vs ~1.45). Repeat customer service contacts signal elevated risk.

Question 5: Call Charge Variations Across Day, Evening, and Night Periods

- **Objective:** Compare average charges across time periods for churned vs. non-churned groups.
- **Excel Formulas:**

Period	Formula for Retained (no)	Formula for Churned (yes)	Delta ($)
Day Charge	`=AVERAGEIF(Data!T2:T4251, "no", Data!I2:I4251)`	`=AVERAGEIF(Data!T2:T4251, "yes", Data!I2:I4251)`	`+$5.37`
Eve Charge	`=AVERAGEIF(Data!T2:T4251, "no", Data!L2:L4251)`	`=AVERAGEIF(Data!T2:T4251, "yes", Data!L2:L4251)`	`+$1.09`
Night Charge	`=AVERAGEIF(Data!T2:T4251, "no", Data!O2:O4251)`	`=AVERAGEIF(Data!T2:T4251, "yes", Data!O2:O4251)`	`+$0.09`

- **Business Insight:** Day charges show the highest discrepancy (+$5.37 per billing cycle for churners). Daytime billing rates are the primary source of billing friction.

Question 6: Relationship Between Total Day Minutes and Churn

- **Objective:** Analyze daytime usage volume differences and correlation with churn.
- **Excel Summary Formulas:**
  - **Pearson Correlation Coefficient (r):**

    ```excel
    =CORREL(Data!G2:G4251, Data!U2:U4251)
    ```
  - **Average Day Minutes (Retained):**

    ```excel
    =AVERAGEIF(Data!T2:T4251, "no", Data!G2:G4251)
    ```
  - **Average Day Minutes (Churned):**

    ```excel
    =AVERAGEIF(Data!T2:T4251, "yes", Data!G2:G4251)
    ```
- **Business Insight:** High daytime callers churn at an elevated rate due to high per-minute costs.

Question 7: Voicemail Plan Impact on Churn Rates

- **Objective:** Contrast churn rates between voicemail subscribers and non-subscribers.

- **Excel Summary Formulas:**
    - **Churn Rate (No Voicemail Plan):**

```excel
=AVERAGEIFS(Data!U2:U4251, Data!E2:E4251, "no")
```

    - **Churn Rate (Active Voicemail Plan):**

```excel
=AVERAGEIFS(Data!U2:U4251, Data!E2:E4251, "yes")
```

- **Business Insight:** Voicemail subscribers churn at roughly half the rate of non-subscribers (~8.54% vs ~16.03%). High feature utilization enhances customer stickiness.

Question 8: International Usage & Charges Comparison

- **Objective:** Compare average international call duration (`total_intl_minutes`) and international charges (`total_intl_charge`).
- **Excel Summary Formulas:**
    - **Average International Minutes (Retained vs. Churned):**

```excel
=AVERAGEIF(Data!T2:T4251, "no", Data!P2:P4251)
```

```excel
=AVERAGEIF(Data!T2:T4251, "yes", Data!P2:P4251)
```

    - **Average International Charges (Retained vs. Churned):**

```excel
=AVERAGEIF(Data!T2:T4251, "no", Data!R2:R4251)
```

```excel
=AVERAGEIF(Data!T2:T4251, "yes", Data!R2:R4251)
```

Question 9: Total Day Calls Correlation with Churn

- **Objective:** Determine if raw daytime call counts (`total_day_calls`) correlate with churn.
- **Excel Correlation Formula:**

```excel
=CORREL(Data!H2:H4251, Data!U2:U4251)
```

(Result $\approx +0.018$, indicating near-zero linear correlation)

- **Business Insight:** Call frequency is independent of churn. Churn is driven by accumulated minute duration and higher rates, not the number of calls initiated.

Question 10: Evening Charges Comparison

- **Objective:** Perform precise mean and median comparisons for evening billing.
- **Excel Summary Formulas:**
  - **Mean Evening Charge (Retained vs. Churned):**

```excel
=AVERAGEIF(Data!T2:T4251, "no", Data!L2:L4251)
```

```excel
=AVERAGEIF(Data!T2:T4251, "yes", Data!L2:L4251)
```

  - **Median Evening Charge (Retained vs. Churned):**

```excel
=MEDIAN(FILTER(Data!L2:L4251, Data!T2:T4251="no"))
```

```excel
=MEDIAN(FILTER(Data!L2:L4251, Data!T2:T4251="yes"))
```

Part 2: Medium-Level Questions (Q1 - Q10)

Question 1: Geographic Churn Analysis by Area Code

- **Objective:** Group churn metrics dynamically by standard area codes (`area_code_408`, `area_code_415`, `area_code_510`).
- **Single Dynamic Array Summary Formula:**

```excel
=LET(
    areas, SORT(UNIQUE(Data!C2:C4251)),
    totals, MAP(areas, LAMBDA(a, COUNTIF(Data!C2:C4251, a))),
    churns, MAP(areas, LAMBDA(a, COUNTIFS(Data!C2:C4251, a,
Data!U2:U4251, 1))),
    rates, churns / totals,
    VSTACK(
        HSTACK("Area Code", "Total Base", "Churned Count", "Churn
Rate"),
        HSTACK(areas, totals, churns, TEXT(rates, "0.00%"))
    )
)
```

Question 2: Support Call Threshold Breakdown

- **Objective:** Segment customers by support call thresholds to find critical risk inflection points.
- **Excel Threshold Table Setup:**

Call Bucket	Criteria Range	Formula for Churn Rate
0 Calls	`=0`	`=AVERAGEIFS(Data!U2:U4251, Data!S2:S4251, 0)`
1 Call	`=1`	`=AVERAGEIFS(Data!U2:U4251, Data!S2:S4251, 1)`
2 Calls	`=2`	`=AVERAGEIFS(Data!U2:U4251, Data!S2:S4251, 2)`
3 Calls	`=3`	`=AVERAGEIFS(Data!U2:U4251, Data!S2:S4251, 3)`

Call Bucket	Criteria Range	Formula for Churn Rate
4 Calls	`=4`	`=AVERAGEIFS(Data!U2:U4251, Data!S2:S4251, 4)`
5+ Calls	`>=5`	`=AVERAGEIFS(Data!U2:U4251, Data!S2:S4251, ">=5")`

- **Critical Takeaway:** Churn jumps sharply from **10.27% at 3 calls** to **45.58% at 4 calls**, reaching **76.19% at 5+ calls**. Interventions must trigger at 3 calls.

Question 3: Account Tenure Quartile Analysis

- **Objective:** Segment account length into quartiles using Excel percentile cutoffs.
- **Quartile Threshold Formulas:**
  - **Q1 Cutoff (25th Percentile):** `=PERCENTILE.INC(Data!B2:B4251, 0.25)`
  - **Q2 Cutoff (50th Percentile):** `=PERCENTILE.INC(Data!B2:B4251, 0.50)`
  - **Q3 Cutoff (75th Percentile):** `=PERCENTILE.INC(Data!B2:B4251, 0.75)`
- **Quartile Churn Rates:**
  - **Quartile 1:** `=AVERAGEIFS(Data!U2:U4251, Data!B2:B4251, "<="&PERCENTILE.INC(Data!B2:B4251, 0.25))`
  - **Quartile 4:** `=AVERAGEIFS(Data!U2:U4251, Data!B2:B4251, ">"&PERCENTILE.INC(Data!B2:B4251, 0.75))`

Question 4: International Plan Billing Friction Analysis

- **Objective:** Analyze churn rates among international plan subscribers split by high vs. low international charges.
- **Excel Formulas (Filter: `international_plan = "yes"`):**
  - **Low Charges (`$< \$2.70$`):**

```excel
=AVERAGEIFS(Data!U2:U4251, Data!D2:D4251, "yes",
Data!R2:R4251, "<2.70")
```

  - **High Charges (`$\ge \$2.70$`):**

```excel
```

```
=AVERAGEIFS(Data!U2:U4251, Data!D2:D4251, "yes",
Data!R2:R4251, ">=2.70")
```

Question 5: Evening Usage Segmentation Analysis

- **Objective:** Categorize evening usage into Low, Medium, and High tiers using percentiles.
- **Excel Tier Formulas:**
  - **Low Evening Usage (**< 166.6 **mins):**

```excel
=AVERAGEIFS(Data!U2:U4251, Data!J2:J4251, "<166.6")
```

  - **High Evening Usage (**> 235.3 **mins):**

```excel
=AVERAGEIFS(Data!U2:U4251, Data!J2:J4251, ">235.3")
```

Question 6: Rule-Based Customer Usage Profile Clustering

- **Objective:** Emulate profile clustering using `IFS` logic across standardized usage.
- **Helper Column Profile Identifier (Column Y):**

```excel
=IFS(
    AND(G2>220, D2="no"), "Heavy Day Caller",
    D2="yes", "International Profile",
    S2>=4, "Support Friction Profile",
    TRUE, "Standard / Balanced User"
)
```

- **Cluster Summary Table:**

```excel
=LET(
    profiles, HSTACK("Heavy Day Caller", "International Profile",
"Support Friction Profile", "Standard / Balanced User"),
    rates, MAP(profiles, LAMBDA(p, AVERAGEIF(Data!Y2:Y4251, p,
Data!U2:U4251))),
```

```
    counts, MAP(profiles, LAMBDA(p, COUNTIF(Data!Y2:Y4251, p))),
    VSTACK(HSTACK("Profile Cluster", "Count", "Churn Rate"),
HSTACK(TRANSPOSE(profiles), TRANSPOSE(counts),
TEXT(TRANSPOSE(rates), "0.00%")))
)
```

Question 7: Voicemail Message Engagement Interaction

- **Objective:** Measure churn rates for voicemail subscribers broken down by message volume categories.
- **Excel Breakdown Formulas:**
    - **Active Plan & 0 Messages Logged:**

```excel
=AVERAGEIFS(Data!U2:U4251, Data!E2:E4251, "yes",
Data!F2:F4251, 0)
```

    - **Active Plan & High Usage (> 20 Messages):**

```excel
=AVERAGEIFS(Data!U2:U4251, Data!E2:E4251, "yes",
Data!F2:F4251, ">20")
```

Question 8: Daytime Inflection Point Identification

- **Objective:** Isolate the exact daytime minute threshold where churn escalates rapidly.
- **Excel Dynamic Binning Formula:**

```excel
=LET(
    bins, HSTACK("0-199 Mins", "200-249 Mins", "250+ Mins"),
    rates, HSTACK(
        AVERAGEIFS(Data!U2:U4251, Data!G2:G4251, "<200"),
        AVERAGEIFS(Data!U2:U4251, Data!G2:G4251, ">=200",
Data!G2:G4251, "<250"),
        AVERAGEIFS(Data!U2:U4251, Data!G2:G4251, ">=250")
    ),
    VSTACK(bins, TEXT(rates, "0.00%"))
```

```
)
```
*(Result: Churn jumps to **58.10%** for usage exceeding 250 day minutes)*

Question 9: International Call Count Interaction by Subscription Plan

- **Objective:** Evaluate how international call volume impacts churn differently based on plan subscription.
- **Excel Matrix Calculations:**
    - **No Plan & Low International Calls ($1 - 3$ calls):**

```excel
=AVERAGEIFS(Data!U2:U4251, Data!D2:D4251, "no",
Data!Q2:Q4251, ">=1", Data!Q2:Q4251, "<=3")
```

    - **Has Plan & Low International Calls ($1 - 3$ calls):**

```excel
=AVERAGEIFS(Data!U2:U4251, Data!D2:D4251, "yes",
Data!Q2:Q4251, ">=1", Data!Q2:Q4251, "<=3")
```

- **Business Insight:** Customers with an international plan who make few calls log a high churn rate (**43.1%**), pointing to subscription fee dissatisfaction among low-volume users.

Question 10: Support Escalation Matrix & Categorization

- **Objective:** Map support call frequency to operational risk tiers.
- **Excel Calculation Table:**

```excel
=LET(
    tiers, SEQUENCE(6, 1, 0, 1),
    rates, MAP(tiers, LAMBDA(t, IF(t=5, AVERAGEIFS(Data!U2:U4251,
Data!S2:S4251, ">=5"), AVERAGEIFS(Data!U2:U4251, Data!S2:S4251,
t)))),
    counts, MAP(tiers, LAMBDA(t, IF(t=5, COUNTIF(Data!S2:S4251,
">=5"), COUNTIF(Data!S2:S4251, t)))),
    HSTACK(tiers, counts, TEXT(rates, "0.00%"))
```

```
)
```

Part 3: Advanced-Level Questions (Q1 - Q5)

Question 1: Excel Additive Risk Scoring Model for Churn Likelihood

- **Objective:** Build an operational, additive point system to predict customer churn probability without external software.

Point Attribution Matrix:

Key Driver Metric	Logic Condition	Point Value	Excel Expression
International Plan	`international_plan = "yes"`	**+3 Points**	`IF(D2="yes", 3, 0)`
Customer Service Calls	`number_customer_service_calls >= 4`	**+4 Points**	`IF(S2>=4, 4, IF(S2=3, 1, 0))`
Day Minutes Usage	`total_day_minutes >= 250`	**+3 Points**	`IF(G2>=250, 3, IF(G2>=200, 1, 0))`
Voicemail Plan	`voice_mail_plan = "no"`	**+1 Point**	`IF(E2="no", 1, 0)`

- **Excel Risk Score Formula (Column Z on Data Sheet):**

```excel
=(IF(D2="yes", 3, 0)) + (IF(S2>=4, 4, IF(S2=3, 1, 0))) +
(IF(G2>=250, 3, IF(G2>=200, 1, 0))) + (IF(E2="no", 1, 0))
```

- **Risk Tier Categorization (Column AA):**

```excel
=IFS(Z2>=6, "High Risk", Z2>=3, "Medium Risk", TRUE, "Low Risk")
```

- **Validation — Empirical Churn Rate by Risk Tier:**
  - **High Risk Tier Churn Rate:** `=AVERAGEIF(Data!AA2:AA4251, "High Risk", Data!U2:U4251)`
  - **Low Risk Tier Churn Rate:** `=AVERAGEIF(Data!AA2:AA4251, "Low Risk", Data!U2:U4251)`

Question 2: Time Series Dynamics and Seasonal Trend Analysis

- **Objective:** Quantify historical trends, period-over-period delta, and linear trajectory.
- **Excel Moving Average & Trend Formulas (Sheet: `TimeSeries_Analysis`):**
    - **3-Period Moving Average Churn Rate:**

```excel
=AVERAGE(C2:C4)
```

    - **Linear Trend Slope (β_1):**

```excel
=SLOPE(ChurnRate_Range, TimePeriod_Range)
```

    - **Seasonality Index:**

```excel
=Monthly_Avg_Churn / Overall_Mean_Churn
```

Question 3: Advanced Behavioral Customer Segmentation

- **Objective:** Segment customers into actionable strategic buckets using combined logical criteria.
- **Excel Multi-Condition Segment Formula (Column AB):**

```excel
=IFS(
    S2>=4, "Segment A: Support Friction Risk",
    AND(D2="yes", R2>=2.70), "Segment B: International Pricing Shock",
    G2>=250, "Segment C: High Day Volume Billing Shock",
    AND(E2="yes", F2>20), "Segment D: Sticky Engaged User",
    TRUE, "Segment E: Low Risk Baseline"
)
```

Strategic Segment Performance Dashboard:

```excel
```

```
=LET(
    seg_names, HSTACK("Segment A: Support Friction Risk", "Segment B:
International Pricing Shock", "Segment C: High Day Volume Billing
Shock", "Segment D: Sticky Engaged User", "Segment E: Low Risk
Baseline"),
    counts, MAP(seg_names, LAMBDA(s, COUNTIF(Data!AB2:AB4251, s))),
    churns, MAP(seg_names, LAMBDA(s, COUNTIFS(Data!AB2:AB4251, s,
Data!U2:U4251, 1))),
    rates, churns / counts,
    VSTACK(
        HSTACK("Strategic Segment Name", "Total Accounts", "Churn
Count", "Segment Churn Rate"),
        HSTACK(TRANSPOSE(seg_names), TRANSPOSE(counts),
TRANSPOSE(churns), TEXT(TRANSPOSE(rates), "0.00%"))
    )
)
```

Question 4: Multivariate Interaction Effect Modeling

- **Objective:** Model the interaction effect between international plan status and international minutes.
- **Excel Multi-Variable Linear Regression Setup:**
    - **Dependent Variable (Y):** Column `U` (`is_churn`)
    - **Independent Variables (X):** International Plan (`V`), Intl Minutes (`P`), and Interaction Term (`V * P`, Column AC).
- **Interaction Term Calculation (Column AC):**

```excel
=V2 * P2
```

- **Regression Analysis via Data Analysis Toolpak or Formulas:**
    - **Interaction Effect Slope (β_3):**

```excel
=INDEX(LINEST(Data!U2:U4251, Data!P2:AC4251), 1, 1)
```

- **Interpretation:** Positive interaction coefficient confirms that additional international minutes increase churn risk far more steeply for plan subscribers than non-subscribers due to unexpected per-minute billing overages.

Question 5: Financial Impact Model & Campaign ROI Calculation

- **Objective:** Calculate overall revenue loss, Customer Lifetime Value (CLV), and program return on investment (ROI).

Financial Input Assumptions (Sheet: `Financial_Model`):

- **Average Monthly Revenue Per User (ARPU):** `$58.50`
- **Gross Margin (M):** `75.0%`
- **Monthly Churn Rate (r):** `1.25%` (equivalent to ~14.07% annual churn)
- **Retention Campaign Cost per Targeted Customer:** `$45.00`
- **Targeted Retention Success Rate:** `30.0%`

Financial Calculations & Formulas:

1. **Customer Lifetime Value (CLV):** `$$\text{CLV} = \frac{\text{ARPU} \times M}{r}$$`

   ```excel
   =(58.50 * 0.75) / 0.0125
   ```
   (Result `$\rightarrow \$3,510.00$` per customer)

2. **Total Annual Gross Revenue Lost to Churn:**

   ```excel
   =COUNTIF(Data!T2:T4251, "yes") * 58.50 * 12
   ```
   (Result `$\rightarrow \$419,796.00$` across sample)

3. **Total Lifetime Value Exposed (CLV Pool):**

   ```excel
   =COUNTIF(Data!T2:T4251, "yes") * 3510
   ```
   (Result `$\rightarrow \$2,098,980.00$`)

4. **Targeted Campaign Cost (High Risk Accounts, Score ≥ 6):**

   ```excel
   =COUNTIF(Data!AA2:AA4251, "High Risk") * 45
   ```

```
```

5. **Net Preserved CLV Revenue:**

```excel
=(COUNTIF(Data!AA2:AA4251, "High Risk") * 0.30) * 3510
```

6. **Net Campaign ROI (%):**

```excel
=(( (COUNTIF(Data!AA2:AA4251, "High Risk") * 0.30) * 3510 ) -
(COUNTIF(Data!AA2:AA4251, "High Risk") * 45)) /
(COUNTIF(Data!AA2:AA4251, "High Risk") * 45)
```

(Format cell as Percentage → Projected Campaign ROI $\approx 2,240\%$)

Executive Implementation Summary & Action Plan

1. **Support Call Circuit Breaker:** Implement automated alerts for accounts logging **3 support calls**; automatically route them to senior retention specialists before the 4th call threshold.
2. **Day Usage Capping:** Offer high-volume daytime callers (>200 minutes) flat-rate unlimited daytime plans to eliminate bill shock.
3. **International Plan Restructuring:** Replace per-minute international overage rates with flat-rate bundled packages.
4. **Voicemail Onboarding Incentive:** Offer complimentary voicemail features during onboarding to increase customer engagement and double retention rates.